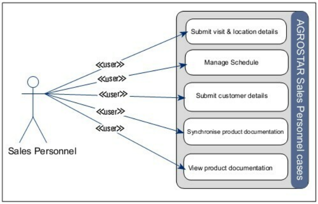

Android devices in the market today.

An overview of device hardware

The device hardware largely consists of a System-on-Chip (SoC) and miscellaneous accessories. A SoC typically consists of the following components:

- **CPU:** In the case of Android, most processors on the SoCs are based on ARM technology.
- **Memory:** Memory is required to perform the various tasks smartphones and tablets are capable of, and therefore SoCs come with various memory architectures on board.
- **GPU:** The graphics processing unit is responsible for handling complex 3D games on the smartphone or tablets. GeForce, Adreno, ARM Mali and Power VR are some of the popular GPU models available in the market.
- **Northbridge:** This component handles communications between the CPU and other components of the SoC, including the Southbridge.
- **Southbridge:** This is a second chipset, usually found on computers, that handles various I/O functions. In some cases, a Southbridge can be found on SoCs too.
- **Cellular radios:** Some SoCs also come with certain modems on board that are needed by mobile operators.
- **Other radios:** Some SoCs may also have other components responsible for other types of connectivity, including Wi-Fi, GPS/GLONASS or Bluetooth. Again, the Snapdragon S4 is a good example.
- Other circuitry

Some of the well-recognised SoCs in the market are described below:

- **NVIDIA Tegra 3:** The Tegra 3 comes with a quad-core CPU, and is currently employed by various Android devices, including, but not limited to, the Asus Eee Transformer Pad, HTC One X (international version), the Asus Transformer Pad 300, the LG Optimus 4X HD and others.
- **Qualcomm Snapdragon S4:** Snapdragon S4 has a dual-core processor that's similar to the ARM Cortex-A15 CPU. It also offers HD video recording and playing support, integrated Adreno GPU capabilities, and packs in a 4G LTE modem, which makes it a preferred choice by many manufacturers planning to launch devices in the US. It also handles Wi-Fi, GPS/GLONASS, and Bluetooth on most devices. Some Android devices that use this SoC include the HTC One S, Asus Transformer Pad Infinity, North-American HTC One X, HTC EVO 4G LTE, Sony Xperia S, North American Samsung Galaxy S3, and others.
- **Samsung Exynos 4 Quad:** Samsung has its own SoC platform, the ARM based Exynos family. The latest addition, the Exynos 4 Quad, packs a 1.4 GHz quad-core ARM Cortex-A9 CPU and the ARM Mali-400 MP4 quad-core GPU. The processor supports 3D gaming, fast multi-tasking, and HD video recording and playback. The Exynos 4 Quad is used in the Galaxy S3 (international version) and in the Meizu MX Quad. Previous Exynos generations can be found in the Galaxy S2, Galaxy Note, Galaxy Tab 7.7, Galaxy Tab 7.0 Plus, Galaxy S, Droid Charge, Exibit 4G, Infuse 4G, and also in non-Samsung devices such as the Meizu MX and Meizu M9.
- **Texas Instruments OMAP 4:** The latest TI OMAP SoCs family is the fourth generation of OMAPs, or OMAP 4, which has a dual-core ARM Cortex-A9 45nm-based architecture. These offer PowerVR graphics. Some Android devices that pack TI OMAP 4 SoCs include Motorola Atrix 2, Motorola Droid 3, Motorola Droid Bionic, Motorola Droid RAZR, Motorola Xyboard, some Samsung Galaxy S2 models, Amazon's Kindle Fire, Samsung Galaxy Tab 2 7.0, Samsung Galaxy Tab 2 10.1, Samsung Galaxy Nexus, LG Optimus 3D and LG Optimus Max.
- **ST-Ericsson NovaThor:** NovaThor SoCs come with a 1 GHz or 1.2 GHz dual-core ARM Cortex-A9 processor, single-core ARM Mali 400 GPUs, and wireless support (GSM/EDGE/HSPA/HSPA+, depending on the model used). Some devices that rely on this SoC include the Sony Xperia P, Sony Xperia U, Sony Xperia Sola, Samsung Galaxy Ace 2, Samsung Galaxy Beam and the HTC Sensation for China.
- **Rockchip:** Rockchip is a series of SoC integrated circuits manufactured by the Fuzhou Rockchip Electronics Company. These integrated circuits are mainly for embedded systems applications in mobile entertainment devices such as smartphones, tablets, e-book readers, set-top boxes, media players, and personal video and MP3 players.
- **AllWinner A1X:** The AllWinner A1X, known under Linux as sunxi, is a family of SoC devices designed by AllWinner. Currently, the family consists of the A13 and the A10. They incorporate the ARM Cortex-A8 as the main processor, and the Mali 400 as the GPU. It is known for its ability to boot from an SD card, in addition to the Android OS usually installed on the flash memory of the device. A10 (sun4i) is the current full-featured SoC, whereas A13 (sun5i) has lower power consumption, and is a lower-cost version of the A10,